Sagnik Bhattacharjee

Email | LinkedIn | GitHub

Education

Indian Institute of Engineering Science and Technology, Shibpur — Kolkata, India
BTech in Information Technology — *Expected July 2029*

Projects

TinyTracer | *Python* | GitHub — December 2025
- Implemented a physically based path tracer with recursive light transport, Monte Carlo sampling, and analytic sphere/quad intersections
- Engineered modular materials (Lambertian, metal, dielectric, emissive) with accurate normals, face orientation detection, and Schlick reflectance
- Designed a multiprocessing rendering pipeline with stochastic anti-aliasing, movable camera optics, and a lightweight scene graph
- Delivered gamma-corrected PPM output via a custom framebuffer and complete ray–object shading pipeline

ChessEngine - TouchGrass | *Python* | GitHub — October 2025
- Built a fully rules-accurate chess backend with legal move generation, state transitions, threefold repetition detection, and a complete undo/redo stack
- Integrated Zobrist hashing for efficient board state encoding, enabling fast repetition tracking and transposition lookups
- Implemented a modular alpha-beta search atop the backend's move execution system, supporting promotions, castling, and en passant
- Designed an extensible API exposing board state, legal moves, move execution, and termination detection for engine or UI clients

dunefetch | *Python, curses* | GitHub — August 2025
- Developed a hybrid terminal application combining neofetch-style system information with an interactive falling-sand physics simulation
- Implemented a cellular-automata particle system with multiple interacting materials and deterministic update rules
- Built real-time terminal visualization using curses with grid-based rendering and responsive keyboard controls
- Structured simulation logic using an object-oriented design for extensible material behavior and physics

amine | *Python, Flask, PyAutoGUI, SQLite* | GitHub | ProductHunt — August 2024 – October 2024
- Reached #10 on ProductHunt with 450+ downloads by building a multi-threaded distraction-blocking productivity app
- Implemented system-level mouse boundary enforcement, keyboard interception, and full-screen tracking to prevent app switching
- Designed an analytics dashboard using Matplotlib, Seaborn, and Plotly with session insights and CSV export
- Engineered SQLite-backed session management, desktop notifications, and input control, gaining 170+ GitHub stars

yars | *Python, requests, urllib3* | GitHub — October 2024
- Bypassed rate limits and anti-bot filters using automated retries, exponential backoff, proxy rotation, and randomized user agents
- Implemented global and subreddit-targeted search with pagination handling, returning structured metadata-rich results
- Built a recursive comment extraction system parsing nested Reddit threads into hierarchical JSON trees
- Developed user activity and subreddit feed scrapers supporting multiple sorting modes with controlled request scheduling

Skills

Languages: Python, C/C++, SQL
Frameworks and Libraries: Flask, FlaskWebGUI, Raylib, curses, PyAutoGUI, pandas, NumPy, Matplotlib, Plotly, Seaborn
Developer Tools: Git, SQLite
Competitive Programming: Codeforces Pupil (1335)